

APPENDIX A

Table 3

Public Data Sources used for this study

Data Source	Description	Purpose	Official URL
Microsoft Learn	Official Microsoft product documentation, tutorials and implementation guidance	Product documentation, architecture, implementation guidance	https://learn.microsoft.com
Microsoft Learn Q&A	Community questions with Microsoft-validated answers	Real enterprise technical questions and accepted answers	https://learn.microsoft.com/answers
Azure Architecture Center	Reference architectures and solution patterns	Enterprise architecture guidance	https://learn.microsoft.com/azure/architecture
Microsoft Cloud Adoption Framework	Cloud migration and governance methodologies	Implementation methodology	https://learn.microsoft.com/azure/cloud-adoption-framework
AWS Documentation	Official AWS product documentation	Product documentation	https://docs.aws.amazon.com
AWS Well-Architected Framework	Architecture best practices	Architecture guidance	https://docs.aws.amazon.com/wellarchitected
AWS Prescriptive Guidance	Implementation patterns and migration guidance	Implementation guidance	https://docs.aws.amazon.com/prescriptive-guidance
AWS re:Post	Enterprise technical questions and accepted answers	Real question–answer pairs	https://repost.aws
Google Cloud Documentation	Official Google Cloud documentation	Product documentation	https://cloud.google.com/docs
Google Cloud Architecture Center	Cloud architecture guidance	Reference architectures	https://cloud.google.com/architecture
Databricks Documentation	Data engineering and AI documentation	Product and implementation guidance	https://docs.databricks.com
Snowflake Documentation	Data platform documentation	Product documentation	https://docs.snowflake.com

Salesforce Documentation	CRM implementation and architecture guidance	Product documentation	https://help.salesforce.com
IBM Documentation	Enterprise architecture and implementation guidance	Product documentation	https://www.ibm.com/docs
Cisco Documentation	Networking and security documentation	Product documentation	https://www.cisco.com/c/en/us/support/docs
Stack Overflow (Filtered)	Enterprise architecture and conceptual discussions only	Supplementary enterprise questions	https://stackoverflow.com

Table 4

Enterprise Knowledge Variables

Variable	Type	Description
Public_Knowledge_Only	Ordinal	Indicates whether the question can be answered using public documentation alone (Yes, Partial, No)
Enterprise_Knowledge_Required	Categorical	Additional knowledge required (Methodology, Pricing, Governance, Proposal Library, SME, etc.)
Suggested_Repository	Categorical	Recommended enterprise repository for answering the question
Human_Intervention	Binary	Indicates whether SME or human review is required
Knowledge_Gap_Category	Categorical	Category of missing enterprise knowledge preventing autonomous AI response

Table 5

Response Quality Variables

Variable	Type	Description
Completeness	Numeric	Percentage of expected answer elements present
Correctness	Numeric	Percentage of facts aligned with authoritative sources
Groundedness	Numeric	Percentage of statements supported by retrieved evidence
Citation_Coverage	Numeric	Percentage of claims supported by authoritative citations
Response_Quality_Index (RQI)	Numeric	Composite score calculated from Completeness, Correctness, Groundedness and Citation Coverage $RQI = 0.25(Completeness) + 0.25(Correctness) + 0.25(Groundedness) + 0.25(Cit$

		ation Coverage)
--	--	-----------------

Figure 2

Screenshots of Data (1 / 2)

	A	B	C	D	E	F	G
1	Question_ID	Vendor	Source Website	Question URL	Source URL	Question	Official Answer
2	EPK-0001	AWS	aws.amazon.com	https://aws.amazon.com/directconnect/faq/s/	https://aws.amazon.com/directconnect/faq/s/	If I associate virtual private gateways (VPGs) to an AWS Direct Connect gateway, can I continue to use all VPC features?	Networking features, such as Elastic File System, Elastic Load Balancing, Application Load Balancer, Security Groups, Access Control List, and AWS PrivateLink, work with AWS Direct Connect gateway.
3	EPK-0002	AWS	aws.amazon.com	https://aws.amazon.com/datasync/faq/s/	https://aws.amazon.com/datasync/faq/s/	Does AWS DataSync support VPC endpoints using AWS PrivateLink?	Yes, VPC endpoints are supported for data movement use cases. You can use VPC endpoints to ensure data transferred between your AWS
4	EPK-0003	AWS	aws.amazon.com	https://aws.amazon.com/disaster-recovery/faq/s/	https://aws.amazon.com/disaster-recovery/faq/s/	How is my data encrypted while in transit from my source server to AWS using AWS PrivateLink?	Your replicated data is encrypted in transit using TLS 1.2, and is transferred directly from your source servers to your VPC. You can
5	EPK-0004	AWS	aws.amazon.com	https://aws.amazon.com/efs/faq/s/	https://aws.amazon.com/efs/faq/s/	How do I control which Amazon EC2 instances can access my file system?	You control which EC2 instances can access your file system using VPC security group rules and IAM policies. Use VPC security groups to
6	EPK-0005	AWS	aws.amazon.com	https://aws.amazon.com/transit-gateway/faq/s/	https://aws.amazon.com/transit-gateway/faq/s/	Can I use the Route Analyzer to analyze security group rules and network ACL rules in VPC?	No, Route Analyzer only verifies Transit Gateway route tables. Security Group rules and Network ACL rules are not a part of the analysis.
7	EPK-0006	AWS	aws.amazon.com	https://aws.amazon.com/shield/faq/s/	https://aws.amazon.com/shield/faq/s/	Which AWS network security services does AWS Shield network security director support?	Currently in preview, AWS Shield supports analysis of AWS WAF, VPC security groups, and VPC network access control lists (network ACLs).
8	EPK-0007	AWS	aws.amazon.com	https://aws.amazon.com/privatelink/faq/s/	https://aws.amazon.com/privatelink/faq/s/	How secure is an AWS PrivateLink connection?	The security of AWS PrivateLink relies on three factors: the path, the policies, and mode of communication. The path between a VPC endpoint and a service trace within AWS and does not traverse the
9	EPK-0008	AWS	aws.amazon.com	https://aws.amazon.com/blogs/mt/announcing-aws-well-architected-migration-lens/	https://aws.amazon.com/blogs/mt/announcing-aws-well-architected-migration-lens/	Who should use the Migration Lens?	All AWS customers designing or implementing a migration to AWS Cloud and searching for guidance regarding best practices and things to consider as part of the move can leverage the Migration Lens. This
10	EPK-0009	AWS	aws.amazon.com	https://aws.amazon.com/datasync/faq/s/	https://aws.amazon.com/datasync/faq/s/	Can I use AWS DataSync with AWS Direct Connect?	You can use AWS DataSync with your Direct Connect link to access public service endpoints or private VPC endpoints . When using VPC endpoints, data transferred between the DataSync agent and AWS
	EPK-0010	AWS	aws.amazon.com	https://aws.amazon.com/datasync/faq/s/	https://aws.amazon.com/datasync/faq/s/	How does AWS DataSync access my Amazon FSx for Lustre file system?	When you create a DataSync task to copy to or from your FSx for Lustre file system, the DataSync service will create Elastic Network Interfaces (ENIs) in the same VPC and subnet as your file system is located

Figure 3

Screenshots of Data (2 / 2)

	A	J	K	L	M	N	O	P	Q	R	S
1	Question_ID	Knowledge Asset Category	Technology Domain	Cloud Platform	Presales Category	Business Capability	Question Intent	Question Type	Question Complexity	Question Length	Number of Technologies Mentioned
2	EPK-0001	Security Guide	Security / Governance	AWS	Solution Architecture	Risk, compliance and identity control	Request presales guidance	Architecture	High	21	4
3	EPK-0002	Migration Guide	Security / Governance	AWS	Migration Assessment	Risk, compliance and identity control	Request presales guidance	Architecture	High	9	5
4	EPK-0003	Security Guide	Security / Governance	AWS	Solution Architecture	Risk, compliance and identity control	Request presales guidance	Architecture	High	17	3
5	EPK-0004	Security Guide	Security / Governance	AWS	Solution Architecture	Risk, compliance and identity control	Determine implementation	Architecture	High	13	4
6	EPK-0005	Security Guide	Security / Governance	AWS	Solution Architecture	Risk, compliance and identity control	Request presales guidance	Architecture	High	17	2
7	EPK-0006	Security Guide	Security / Governance	AWS	Solution Architecture	Risk, compliance and identity control	Evaluate options / fit	Architecture	High	12	4
8	EPK-0007	Security Guide	Security / Governance	AWS	Solution Architecture	Risk, compliance and identity control	Request presales guidance	Architecture	High	7	3
9	EPK-0008	Well Architected	Security / Governance	AWS	Migration Assessment	Risk, compliance and identity control	Evaluate options / fit	Best Practice	High	6	1
10	EPK-0009	Migration Guide	Security / Governance	AWS	Migration Assessment	Risk, compliance and identity control	Request presales guidance	Architecture	High	9	4
	EPK-0010	Migration Guide	Security / Governance	AWS	Migration Assessment	Risk, compliance and identity control	Determine implementation	Best Practice	High	12	3

Table 6

Proposed Methods

RQ1 Evaluation Metrics	Statistical Techniques: Binary Logistic Regression and Chi-square Test of Independence Machine Learning: Random Forest, XGBoost , BERT-based Text Classification Evaluation Metrics: Classification Accuracy=TP+TN / TP+TN+FP+FN Precision, P =TP/TP+FP, Recall, R =TP/TP+FN F1=2PR/P+R
RQ2 Evaluation Metrics	Statistical Techniques: Chi-square Test, Correspondence Analysis, Association Rule Mining Machine Learning: Multi-label Classification, Graph-based Knowledge Routing Learning-to-Rank Models Evaluation Metrics: Cramér's $V = \sqrt{\chi^2/n(k - 1)}$ Support(A) = Count(A)/N Confidence(A→B)=Support(A∩B)/Support(A) Lift=Confidence/Support(B) Hamming Loss (multi-label) HL=1/NL*($\sum I(y_i \neq \hat{y}_i)$)
RQ3 Evaluation Metrics	Statistical Techniques: Multiple Linear Regression, ANOVA, Pearson Correlation Machine Learning: Retrieval-Augmented Generation, LLM-as-a-Judge (supported by reference-based evaluation), Semantic Similarity Models Response Quality Index: RQI=0.25(COMP)+0.25(CORR)+0.25(GRO)+0.25(CITE), where COMP = Completeness, CORR = Correctness, GRO = Groundedness, CITE = Citation Coverage Evaluation Metrics: Coefficient of Determination R²=1-SSres/SStot Root Mean Square Error RMSE=$\sqrt{\sum (y - \hat{y})^2/n}$ Mean Absolute Error MAE=$\sum y - \hat{y} /n$
RQ4 Evaluation Metrics	Statistical Techniques: Cluster Analysis, Chi-square Test, Pareto Analysis, Association Rule Mining Machine Learning: Clustering, Topic Modelling, Knowledge Graph Analysis Evaluation Metrics: Silhouette Score: S=b-a/max(a,b) Davies–Bouldin Index DB=1/n$\sum \max(\sigma_i + \sigma_j d(c_i, c_j))$ Pareto Contribution Contribution_i=(Frequency_i/∑Frequency)×100